# Medical STT — SOAP Extraction

## 3-Model Performance Comparison

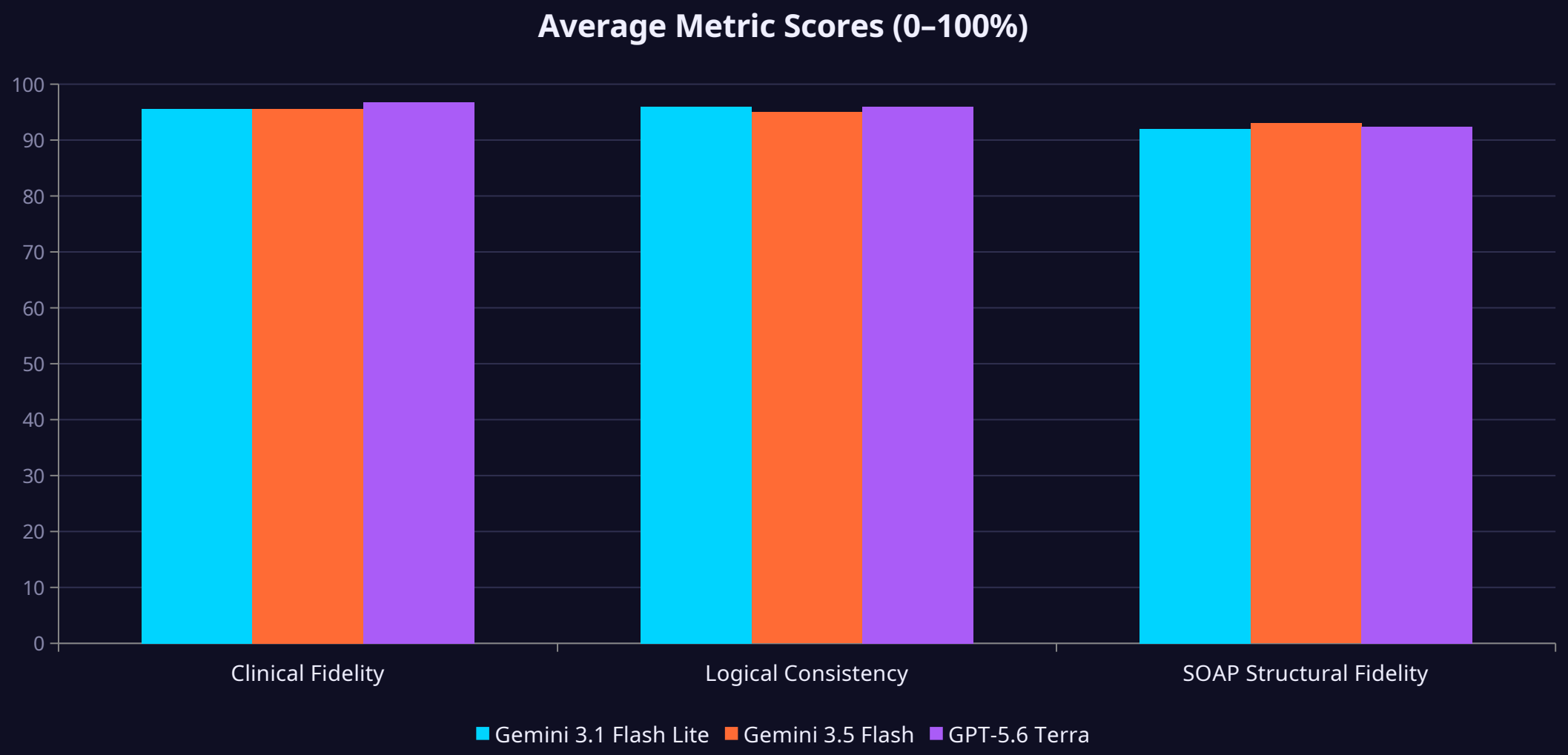
Gemini 3.1 Flash Lite · Gemini 3.5 Flash · GPT-5.6 Terra

5 Clinical Conversations | 3 Metrics | Judge: gemini-3.5-flash-lite

Metrics: Clinical Fidelity · Logical Consistency · SOAP Structural Fidelity (0–100%)

# Overall Model Performance

Average scores across all 5 clinical conversations  (0–100%)



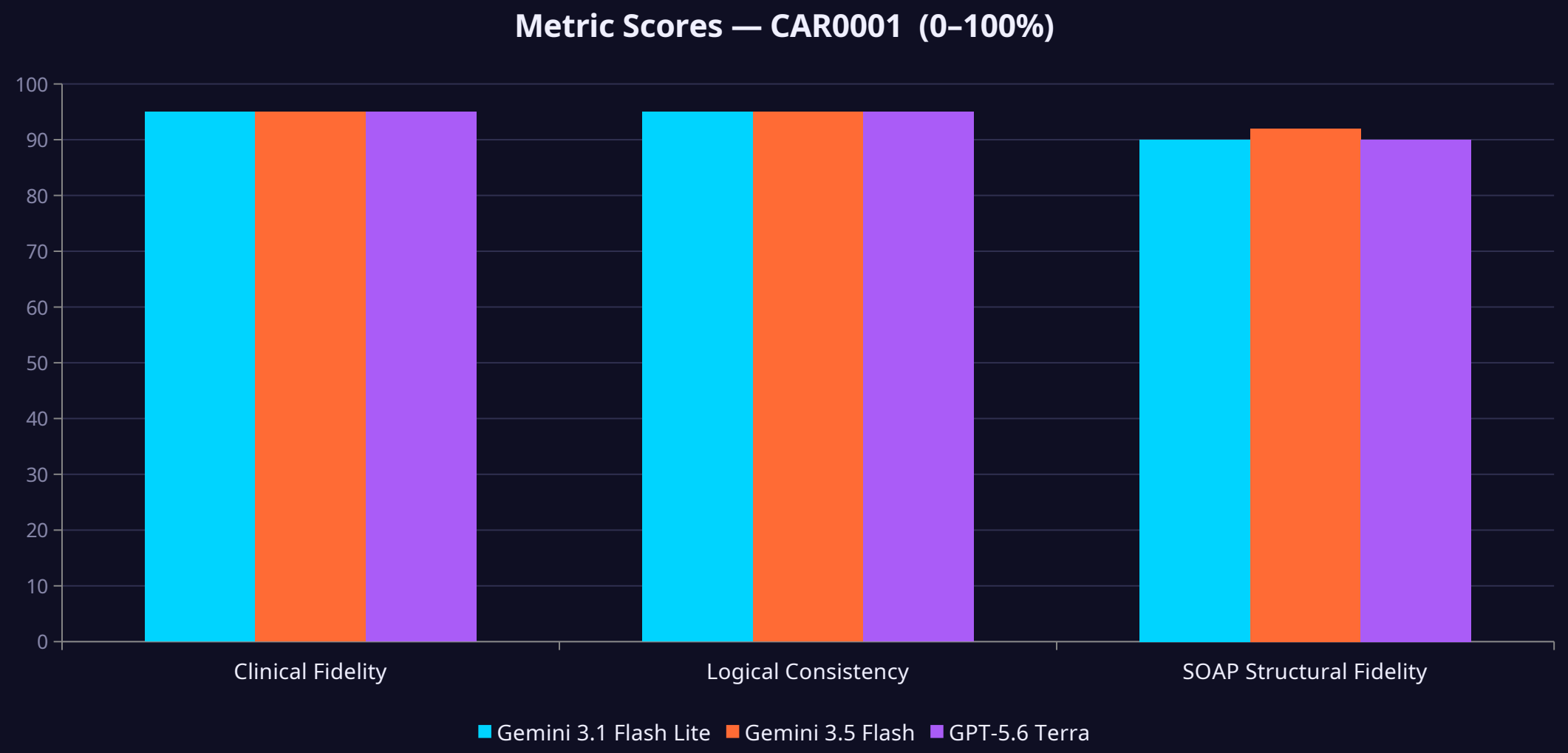
# Full Results — All 15 Evaluations

5 cases × 3 models | Metric scores 0–100% | Judge: gemini-3.5-flash-lite

| Case | Model | Clinical Fidelity | Logical Consistency | SOAP Structural Fidelity | Avg |
|---|---|---|---|---|---|
| CAR0001 | Gemini 3.1 Flash Lite | 95% | 95% | 90% | **93.3%** |
| CAR0001 | Gemini 3.5 Flash | 95% | 95% | 92% | **94.0%** |
| CAR0001 | GPT-5.6 Terra | 95% | 95% | 90% | **93.3%** |
| DER0001 | Gemini 3.1 Flash Lite | 95% | 95% | 90% | **93.3%** |
| DER0001 | Gemini 3.5 Flash | 95% | 95% | 95% | **95.0%** |
| DER0001 | GPT-5.6 Terra | 98% | 95% | 92% | **95.0%** |
| GAS0003 | Gemini 3.1 Flash Lite | 95% | 95% | 90% | **93.3%** |
| GAS0003 | Gemini 3.5 Flash | 95% | 95% | 90% | **93.3%** |
| GAS0003 | GPT-5.6 Terra | 95% | 95% | 90% | **93.3%** |
| MSK0005 | Gemini 3.1 Flash Lite | 95% | 95% | 95% | **95.0%** |
| MSK0005 | Gemini 3.5 Flash | 98% | 95% | 96% | **96.3%** |
| MSK0005 | GPT-5.6 Terra | 98% | 100% | 95% | **97.7%** |
| RES0050 | Gemini 3.1 Flash Lite | 98% | 100% | 95% | **97.7%** |
| RES0050 | Gemini 3.5 Flash | 95% | 95% | 92% | **94.0%** |
| RES0050 | GPT-5.6 Terra | 98% | 95% | 95% | **96.0%** |

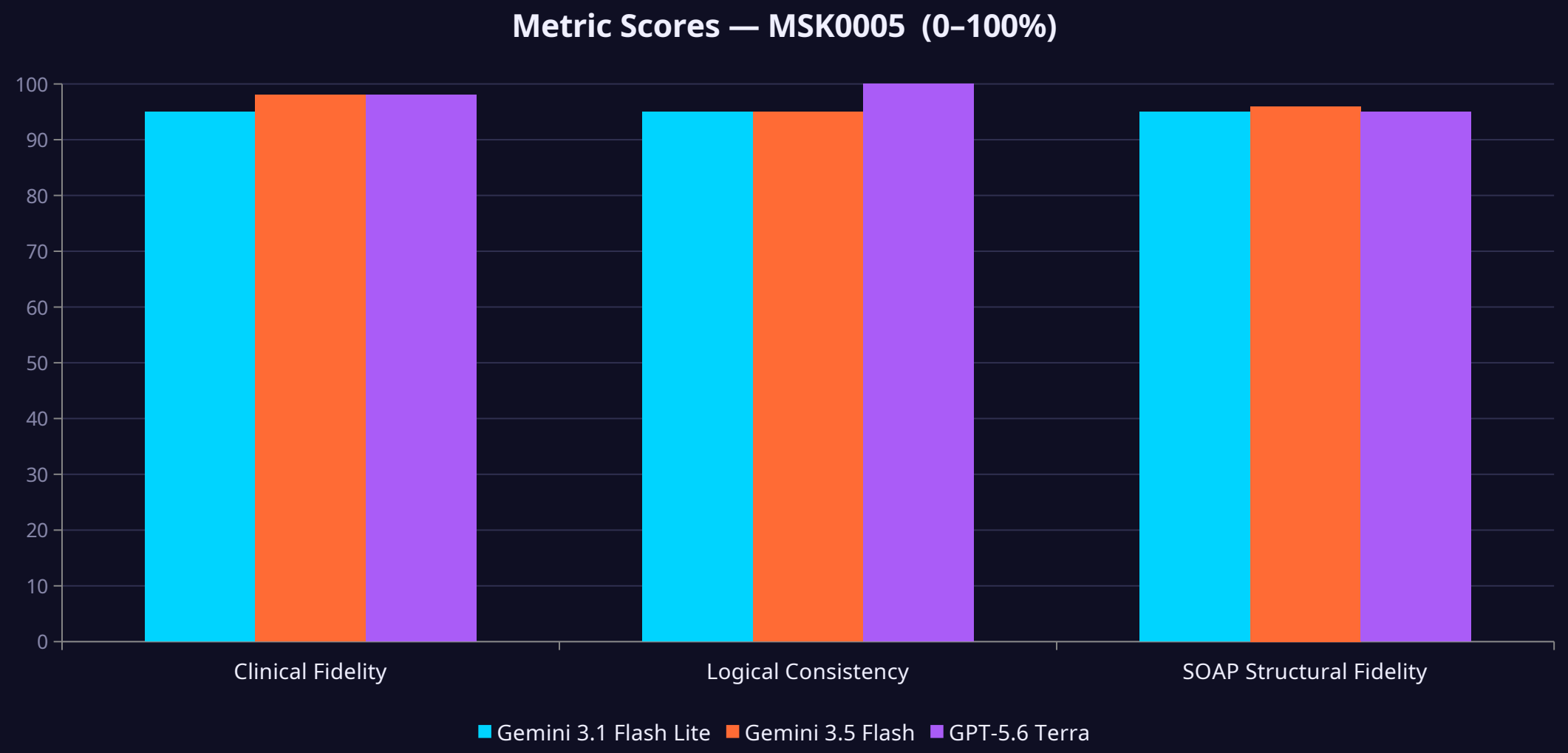
# Case: Cardiology — Chest Pain

Conversation CAR0001 | Performance metrics 0–100%



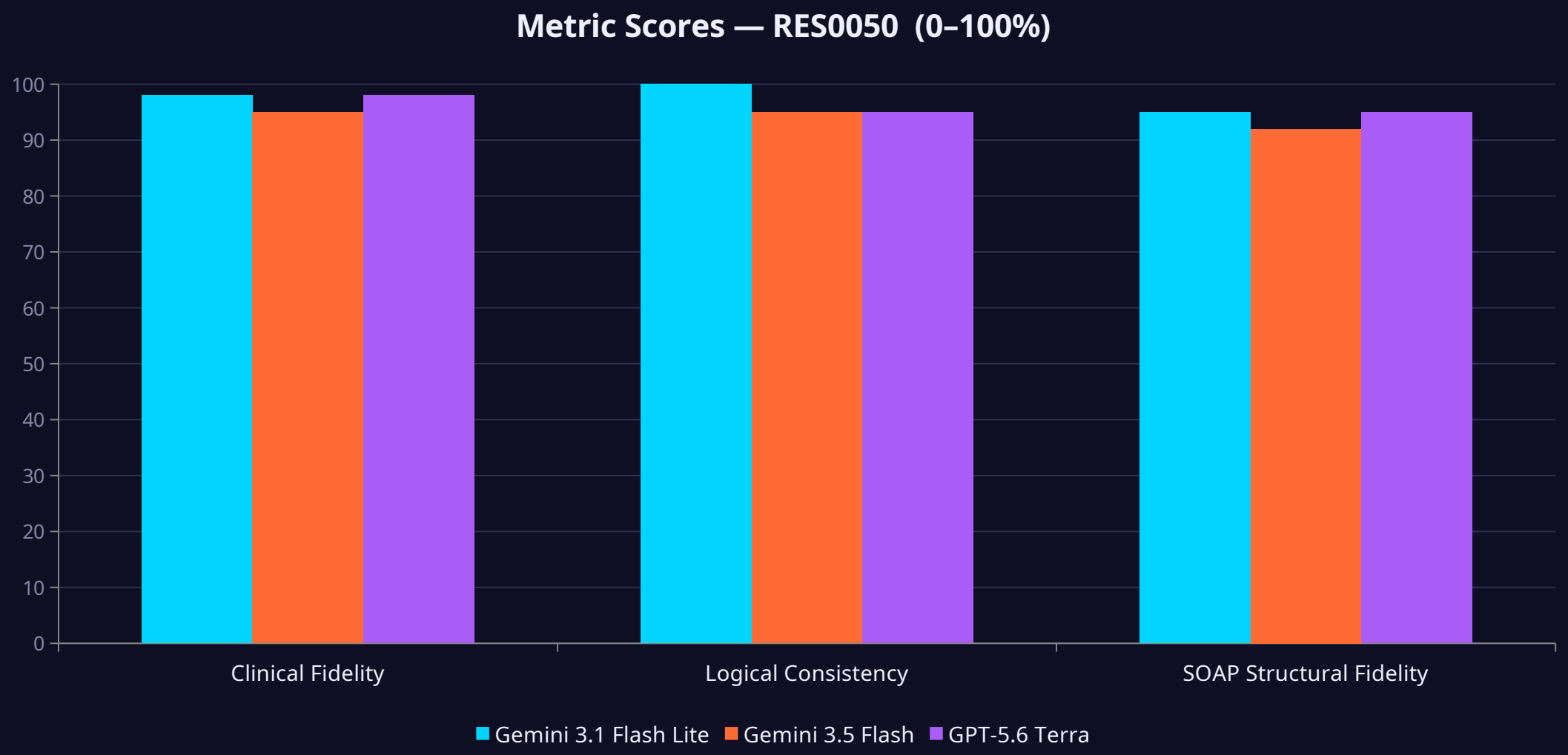
# Case: Musculoskeletal — Elbow Pain

Conversation MSK0005 | Performance metrics 0–100%



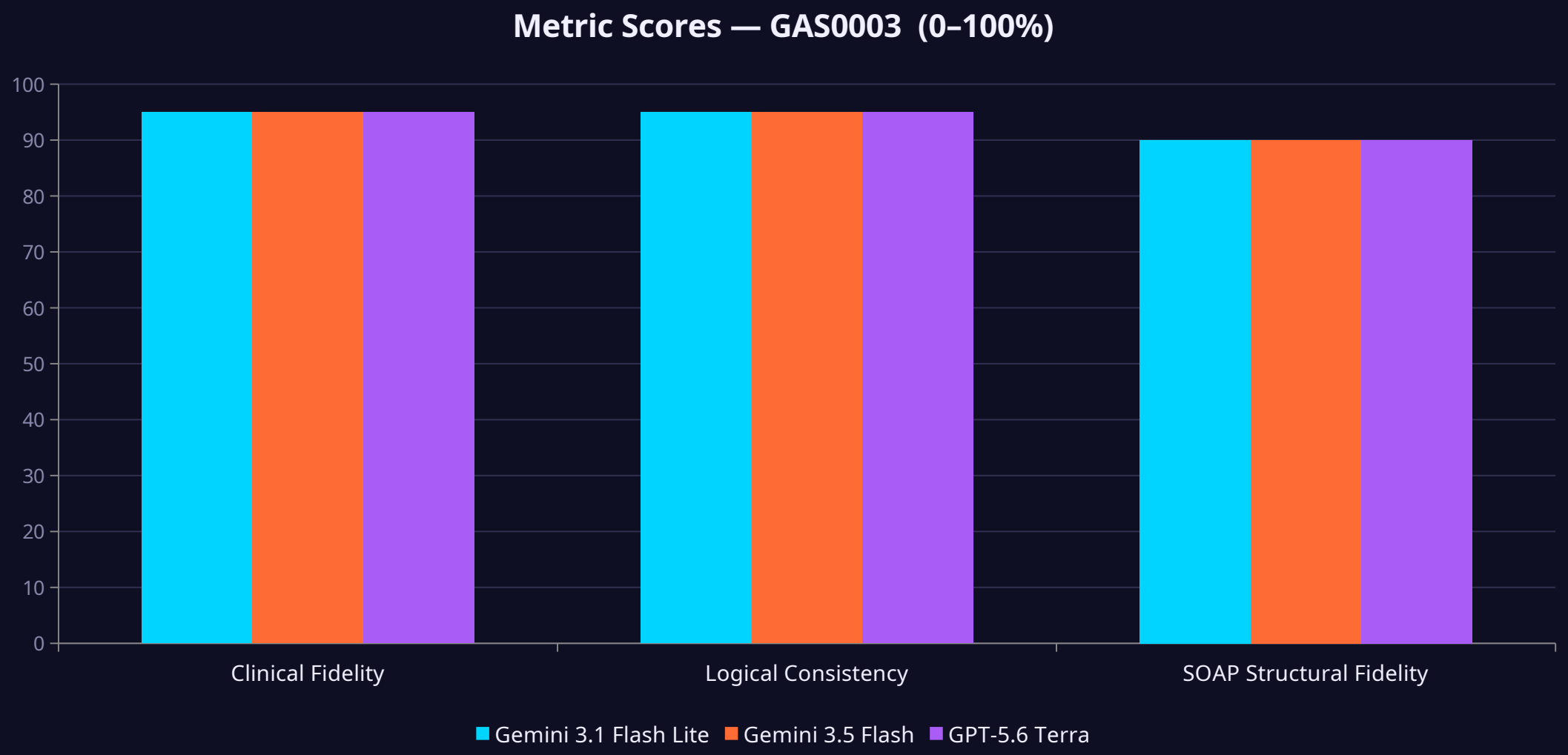
# Case: Respiratory — Chronic Cough

Conversation RES0050 | Performance metrics 0–100%



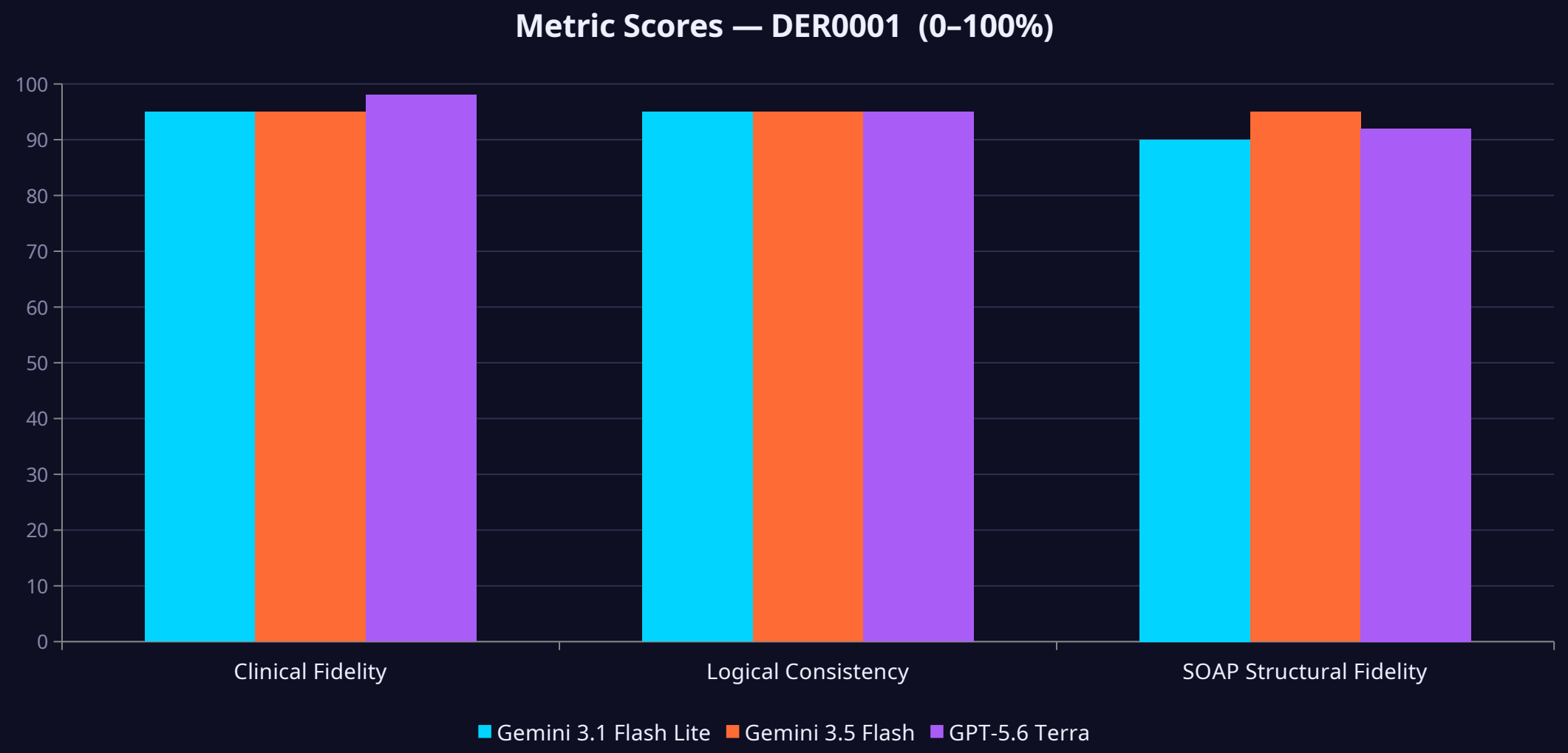
# Case: Gastroenterology — Abdominal Pain

Conversation GAS0003 | Performance metrics 0–100%



# Case: Dermatology — Skin Rash

Conversation DER0001 | Performance metrics 0–100%



# Model Performance Summary

Average scores across all 5 cases (0–100%) | Paper-aligned metrics

| Model | Clinical Fidelity | Logical Consistency | SOAP Structural Fidelity | Overall Avg |
|---|---|---|---|---|
| **GPT-5.6 Terra** | 96.8% | 96.0% | 92.4% | **95.1%** |
| **Gemini 3.1 Flash Lite** | 95.6% | 96.0% | 92.0% | **94.5%** |
| **Gemini 3.5 Flash** | 95.6% | 95.0% | 93.0% | **94.5%** |

🏆 **Highest overall average: GPT-5.6 Terra (95.1%)**

Note: All three models perform exceptionally well (>94% average), demonstrating that modern LLMs are highly capable at structured SOAP extraction from STT transcripts.